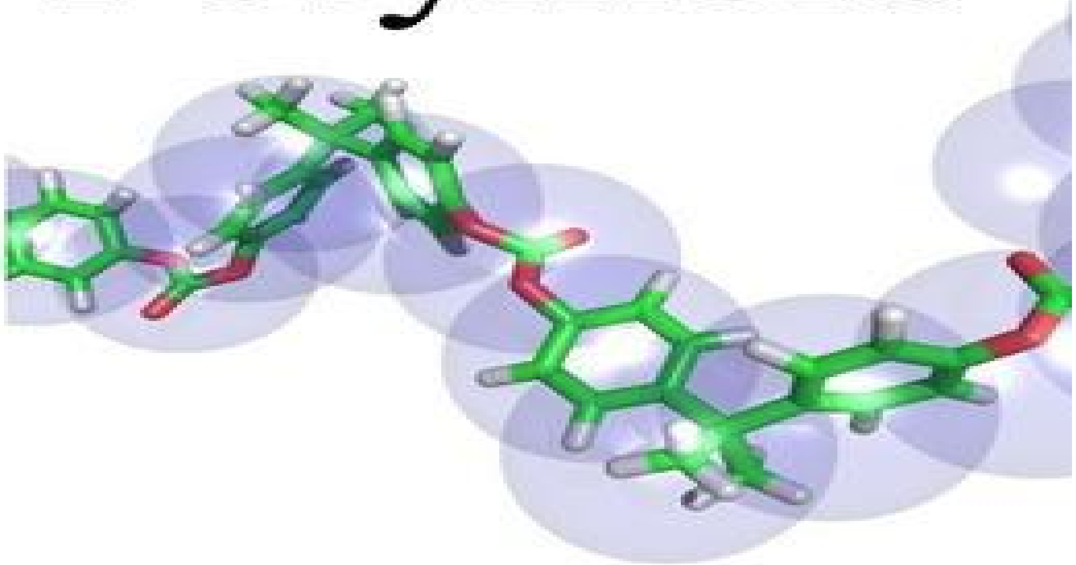
2.1 and 2.2

Naima and Radwa

Nelson Chemistry 12 textbook

Polymers



Introducing to Polymers!

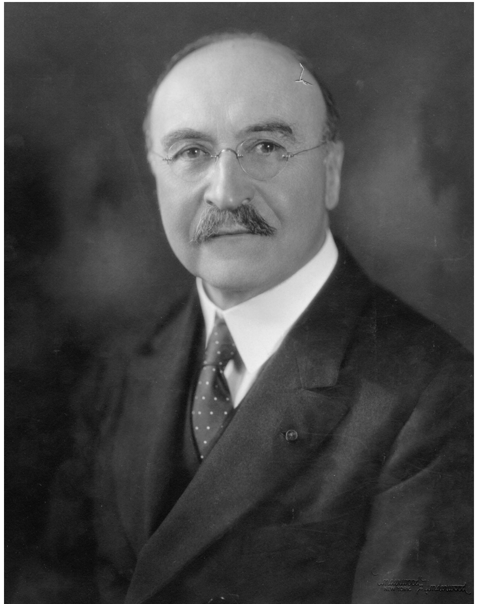
<u>definition</u>

- Polymers - a large usually chain like molecule that is built from small molecules.
- Monomer - one of the repeating small molecules that make polymers
- Homopolymer - a polymer of a single type of monomer.
- Copolymer - a polymer made of two or more different types of monomers combined.

Who Invented Polymers??

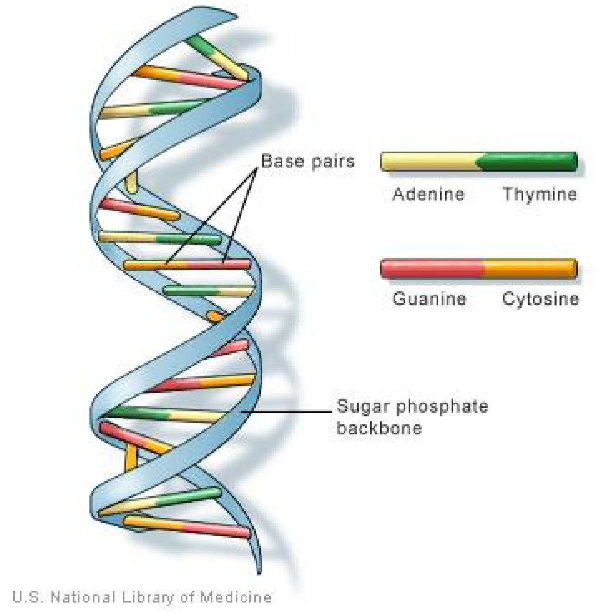
Leo Baekeland

In 1907 **Leo Baekeland** invented the first synthetic polymer, a thermosetting phenol–formaldehyde resin called Bakelite. Despite significant advances in polymer synthesis, the molecular nature of polymers was not understood until the work of **Hermann Staudinger** in 1922.



Example of Polymer

DNA - Dna is a polymer. The monomers called nucleotides are each made up of a base, phosphate and a sugar. There are billions of unique strands of DNA.



2.2 Synthetic Addition Polymers

- Addition polymerization: a reaction in which unsaturated monomers combine with each other to form a polymer.

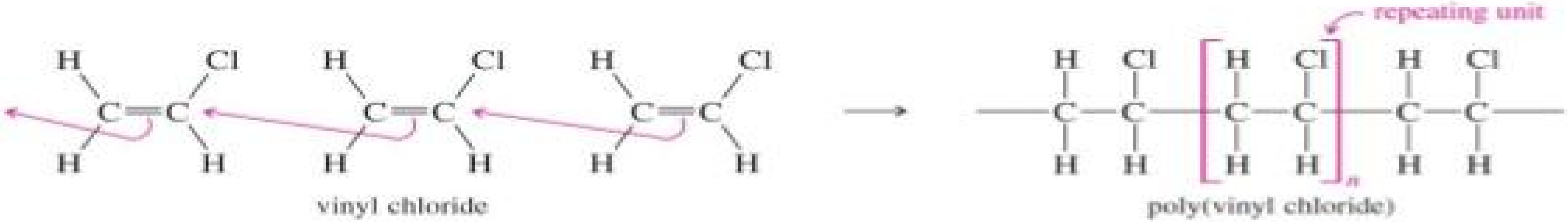
- A reaction where monomers with double bonds are joined together through multiple addition reactions to form a polymer

Addition Polymers

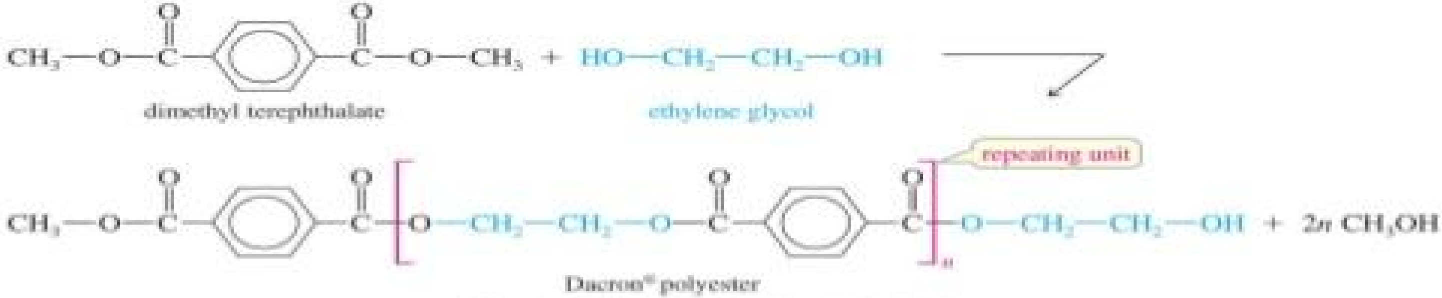
- Three kinds of intermediates:
  - Free radicals
  - Carbocations
  - Carbanions
- Examples of addition polymers:
  - Polypropylene plastics
  - Polystyrene foam insulation
  - Poly(acrylonitrile), Orlon® fiber
  - Poly(methyl α-methacrylate), Plexiglas®

Classes of Polymers

- Addition, or chain-growth, polymers.



- Condensation, or step-growth, polymers.



Synthetic Addition Polymers

- **Addition Polymerization:** a polymerization in which monomer units with **unsaturated carbon-carbon bonds** are joined together without loss of atoms. For example:

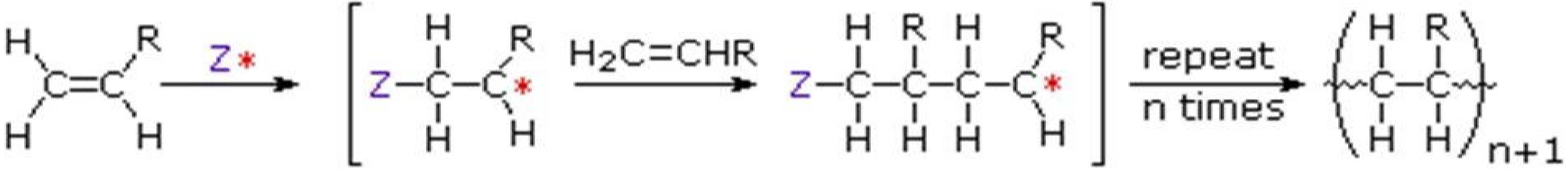
$$\underset{\text{Ethylene}}{nCH_2{=}CH_2} \xrightarrow{\text{catalyst}} \underset{\text{Polyethylene}}{\left(CH_2CH_2\right)_n}$$

- The properties of polymers are dependent on the type of functional group attached to the monomer.

ADDITION POLYMERIZATION

The free radical mechanism can be divided into three stages:

a) **Initiation**
b) **Propagation**
c) **Termination**



Z* is an initiating species

* may be a radical, a cation or an anion

Plastics – Special Addition Polymers

- ***Plastic***: synthetic substance that can be moulded, generally under heat and pressure, that retains its given shape
- Chemically unreactive - stable single bonds
- Flexible and mouldable solids or viscous liquids - van der Waals forces of attraction that exist within them
- Soft & flexible when heated as heat increases molecular motion and allows chains to slide past one another
- Cause major environmental issues
  - http://www.youtube.com/watch?v=GLgh9h2ePYw